

MaintenAI - Intelligent Facility Maintenance & Repair Prioritization Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives open maintenance complaints, equipment records, recent sensor readings, location and occupancy information, technician availability, and spare-part status. At the start, the requests have not yet been assessed or prioritized.
Actions	Read and organize maintenance requests; analyze equipment history and sensor data; estimate safety and operational risk; rank work orders; predict possible failure; assign a suitable technician; schedule preventive maintenance; escalate critical issues; send status updates; and revise priorities after feedback.
Transition Model	Whenever a new complaint, sensor reading, repair update, technician status, or spare-part change is received, the agent updates the equipment condition, risk score, priority ranking, technician assignment, and maintenance schedule.
Path Cost	The cost can be measured through equipment downtime, safety exposure, repair expense, technician effort and travel, service delay, and disruption to users. The objective is to reduce total operational impact while addressing high-risk problems first.
Goal Test	Critical maintenance issues are identified and handled promptly, work orders are prioritized according to risk and impact, preventive tasks are scheduled before likely failure, available resources are used efficiently, and users receive clear status updates.
State Space	All possible combinations of equipment condition, sensor values, open work orders, risk and priority scores, location, occupancy, technician availability, spare-part availability, repair history, and maintenance schedules.
Search Space	All possible sequences of receiving data, assessing condition, calculating risk, ranking requests, assigning resources, scheduling work, completing repairs, monitoring results, and updating future decisions from feedback.
Agents	Single agent. One intelligent maintenance agent coordinates prioritization, scheduling, assignment, escalation, and feedback within the defined facility-management task.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent can only use reported complaints, available records, and sensor readings. Hidden internal faults, incomplete reports, or conditions not covered by sensors may remain unknown until inspection.
Determinism	Stochastic / Non-deterministic	The result of a maintenance decision is not fully predictable. Equipment may deteriorate unexpectedly, repairs may take longer than expected, spare parts may be delayed, or new faults may appear.
Episodes	Sequential	Each decision affects later states. Delaying one repair can increase failure risk, while completing preventive maintenance can reduce future faults and change the remaining workload.
Change	Dynamic	The facility keeps changing while the agent is operating. New complaints can arrive, sensor values can change, technicians can become unavailable, and occupied areas can change during the day.
Values	Discrete & Continuous	Work-order types, rooms, technicians, and equipment categories are discrete. Temperature, vibration, age, time, risk scores, failure probability, and operating measurements can be continuous.
Agents	Single Agent	One intelligent maintenance agent performs the defined prioritization and resource-allocation task. Technicians and users are treated as people and resources in the environment rather than competing AI agents.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through many changing states: a new issue is received, assessed, ranked, assigned, scheduled, repaired, verified, and then monitored. New complaints and feedback continuously create updated states and priorities.

Final Answer: The **MaintenAI Intelligent Facility Maintenance & Repair Prioritization Agent** is best classified as a **Multi-state agent** because it moves through multiple changing states while receiving maintenance data, assessing risk, prioritizing requests, assigning technicians, scheduling repairs, and updating decisions from feedback. Its environment is also **partially observable** and **stochastic** because the agent cannot directly observe every hidden equipment condition or guarantee the exact outcome of a repair or future failure.